

Dipole Antenna Kit

<https://www.rtl-sdr.com/using-our-new-dipole-antenna-kit/>

- 2x 23cm to 1 m telescopic antennas. The telescopic dipoles are detachable from the dipole base via a M5 thread which allows for greater portability and the ability to swap them out. These long telescopic antennas cover VHF to UHF.
- 2x 5cm to 13cm telescopic antennas. These smaller antennas cover UHF to 1090 MHz ADS-B, and even still work decently up to L-band 1.5 GHz frequencies.
- If you mismatch vertical and horizontal polarization or RHCP and LHCP you'll get an instant 20dB loss. If you mismatch vertical/RHCP, vertical/LHCP, horizontal/RHCP, horizontal/LHCP you'll only get a 3dB loss.
 - <https://www.everythingrf.com/community/what-is-cross-polarization>



- shitty ass crimp wtf



- solder blodge a bit, coax shield somehow doesn't want to be soldered
- centre conductor now should be solid
- the usb loose af, maybe can replace with another one

Polarisation

- For vertical polarization, in theory it does not matter which way around you orient the antenna as long as it's vertical. However in practice, you may get slightly better results by having the element connected to the center coax conductor pointing UP.
- For example for LEO VHF satellites you can use a V-dipole configuration. You can also make a somewhat directional antenna by using a bent dipole configuration. Some more examples of dipole configurations can be found on [KK4OBI's page on bent dipoles](#).

Adjusting for Length

- Like with the whip you can use an online calculator to calculate the optimal length for your frequency of interest. We recommend this [dipole calculator](#).
- The exact length does not matter too much, but try to get the lengths as close to what the calculator says as you can. With the dipole you want both elements to be the same length.
- In reality extending the antenna to almost any random length will work just fine for most strong signals. But if you're really trying to optimize those weak signals you'll want to fine tune the lengths.

Balun??



- is this a balun or just some RFI thing???